MILITARY SPECIFICATION

GRAINS, TUBULAR, PYROTECHNIC

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the minimum requirements for tubular pyrotechnic igniter grains for the Mark 265 Igniter.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government Inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 105), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION

NAVORD OD 9375	Standard Methods and Procedures for the Determination of Heat of Explosion of Rocket Propellant Powders

FSC 1336

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DRAWING

Department of the Navy, Bureau of Naval Weapons

2516384	Grain, Pyrotechnic, Igniter

(Requests for the above listed publications should be addressed to the Commanding Officer, Central Technical Documents Office, Louisville 14, Kentucky.)

2.2 Other Publications. -The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Interstate Commerce Commission

49 CFR 71-90	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Request copies from the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction. -The preproduction test sample shall be manufactured using the process methods and equipment proposed for production. Any significant changes after preproduction approval must be approved by the procuring activity prior to incorporation into production. It shall be the responsibility of the contractor to inform the procuring activity of proposed changes after approval of the preproduction sample. In the event of significant changes in the process methods and equipment which in the opinion of the procuring activity may adversely affect the characteristics of the material, additional preproduction samples may be required.

3.2 Production Requirements.

3.2.1 Composition. -The chemical composition of the grains shall be in accordance with Table I.

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TABLE I

COMPOSITION

Ingredients	Limits, % by Weight	
	Minimum	Maximum
Barium Nitrate	15.9	16.9
Potassium Perchlorate	25.2	27.2
Nickel Zirconium Alloy (Zr-Ni) 50/50	51.4	55.4
Ethyl Cellulose	2.5	3.5
Polytetrafluoroethylene (TFE)	0.8	1.2
Moisture Content		0.3

3.2.2 Heat of Explosion. -Heat of explosion shall be 1600 ± 400 BTU/lb.

3.2.3 Density. -Average pellet density shall be 0.130 to 0.140 pound per cubic inch.

3.2.4 Configuration. -The configuration of the grains shall be in accordance with BuWeps drawing 2516384.

3.3 Workmanship. -The material furnished under this specification shall be a uniform product free from foreign materials. It shall be uniform in quality and manufactured in accordance with standard manufacturing procedures of the industry.

3.4 Data Requirements. -No data is required by this specification or by applicable documents referenced in section 2 unless specified in the contract or order (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection record of examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of Tests.

(a) Preproduction Tests (4.4)

(b) Quality Conformance Inspection (4.5)

4.3 Lot Size. -A lot shall consist of material produced by one manufacturer in one continuous operation employing not more than one lot of each ingredient and with no change in formulation or process. If manufacture is by batch process, each batch not exceeding 40 pounds shall constitute a lot.

4.4 Preproduction Tests. -Preproduction tests shall consist of all of the Quality Conformance Inspections and requirements of this specification.

4.5 Quality Conformance Inspections. -The following procedures shall be used for determining that the requirements of 3.2 of this specification have been met.

4.5.1 Sampling. -A minimum of 10 grams of the finished tubular grains shall be drawn from each of five widely-separated places from each lot produced. The entire test sample except two tubular grains shall be cut, shaved, or otherwise reduced in size and thoroughly mixed such that any sub-sample withdrawn for acceptance tests will be representative of the entire test sample. The two uncut grains shall be used for the determination of density as described in 4.5.4.

4.5.2 Chemical Composition. -The composition of each lot of material shall be determined by the following methods. Duplicate determinations for each constituent shall be required. Failure of any determination to comply with the requirements of 3.2.1 shall be cause for rejection of the lot.

4.5.2.1 Preparation of Sample for Extraction.

4.5.2.1.1 Apparatus.

(a) Sochlet Extraction Apparatus (250 ml capacity)

(b) Sochlet Thimble

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(c) Heating Mantle

(d) Glass Evaporating Dish

(e) 250 ml Volumetric Flask

4.5.2.1.2 Procedures.

(a) Weigh a 5 gram sample into a glass evaporating dish and cover with distilled water.

(b) Transfer to a Soxhlet thimble and extract with 150-200 ml distilled water for three hours.

(c) When cool, the solution is transferred to a 250 ml volumetric flask and made up to volume.

4.5.2.2 Determination of Barium Nitrate.

4.5.2.2.1 Apparatus.

(a) 250 ml beaker

(b) Hot plate

(c) 50 ml pipette

(d) Sintered Glass Gooch crucibles, F porosity

4.5.2.2.2 Reagents. -An ammonium carbonate solution shall be made up as follows:

(a) 250 grams $(NH_4)_2 CO_3$, weighed to nearest 0.1 gram

(b) 1000 ml distilled water

(c) 100 ml concentrated NH_4OH

4.5.2.2.3 Procedure.

(a) Transfer 50 ml aliquot portion of 250 ml sample extract to a 250 ml beaker.

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(b) Add 5 ml of ammonium carbonate solution.

(c) Heat with occasional stirring, to 70°C; cool; filter into tared crucible; dry at 100°C overnight and weigh to nearest 0.1 milligram (mg).

(d) Calculation:

$$\% \ Ba(NO_3)_2 = \frac{(\text{wt. } BaCO_3)\ (5)\ (100)\ (1.3243)}{\text{sample wt. } (100 - \% \text{ moisture})}$$

4.5.2.3 Determination of Potassium Perchlorate.

4.5.2.3.1 Apparatus.

(a) 50 ml pipette

(b) Evaporating dish

(c) Steam bath

4.5.2.3.2 Procedure.

(a) Transfer 50 ml aliquot portion of 250 ml sample extract to weighed evaporating dish.

(b) Evaporate to dryness on steam bath and dry at 70°C for 1/2 hour; cool and weigh.

(c) Calculation:

$$\% \ KCLO_4 = \frac{(\text{wt. residue})\ (5)\ (100)}{\text{sample wt. } (100 - \% \text{ moisture})} - \% \ Ba(NO_3)_2$$

4.5.2.4 Determination of Moisture Content.

4.5.2.4.1 Apparatus.

(a) Moisture weighing bottle (approx. 2" dia. x 1") with ground glass stopper.

(b) Desiccator

4.5.2.4.2 Procedure.

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(a) Determine weight of bottle and stopper to nearest 0.1 mg.

(b) Place sample (approx. 2 gms) in bottle and replace stopper. Weigh to nearest 0.1 mg.

(c) Subtract the weight of bottle and stopper, from total wt. of sample, bottle, and stopper to determine true sample weight.

(d) Remove stopper and place in 100°C oven for 2 hours.

(e) Replace stopper, place in desiccator until sample reaches room temperature and weigh to nearest 0.1 mg.

(f) Calculation:

$$\% \text{ moisture} = \frac{\text{Loss in wt} \times 100}{\text{wt of sample}}$$

4.5.2.5 Zirconium Nickel Alloy, Ethyl Cellulose, and Polytetrafluoroethylene (TFE). -The vendor shall be required to certify that the weight percentages for each of these materials are in conformance with the chemical composition as set forth in Table I, paragraph 3.2.1 (see 6.2).

4.6.3 Determination of Heat of Explosion. -Duplicate heat of explosion determinations shall be performed in accordance with NAVORD OD 9375. Failure to comply with the requirements of 3.2.2 shall be cause for rejection of the lot.

4.6.4 Determination of Density. -Two finished tubular grains shall be used to determine the density. Each grain shall be weighed and then checked for density by such approved means as (a) determination of loss of weight in a liquid medium or (b) determination of weight and volume. Castor oil shall be the required liquid medium. Failure to comply with 3.2.3 shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. -Preservation and packaging shall be level C or as specified (see 6.2).

5.1.1 Level C. -Preservation and packaging shall be in accordance with standard commercial practice to afford protection against damage, contamination of the product by moisture, corrosion, or other foreign matter.

5.2 Packing. -Packing shall be level C or as specified (see 6.2).

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5.2.1 Level C. -The product shall be shipped in commercial type containers complying with ICC Code of Federal Regulations 49 CFR 71-78, and so constructed to withstand normal transportation and handling abuse and must be suitably sealed to prevent contamination of the contents by moisture or foreign matter.

5.3 Marking. -Unless otherwise specified by the contract or order, unit packages, imtermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and shall include but not be limited to the following (see 6.2):

(a) Manufacturer's name and address

(b) Gross and net weights

(c) Lot number and date of manufacture

(d) The number of this specification

(e) Safety precautions or unusual storage requirements

(f) Purchase order or contract number

6. NOTES

6.1 Intended Use. -The material purchased in accordance with this specification is intended for use as the main charge in the Igniter Mark 265.

6.2 Ordering Data. -Procurement documents shall specify but not be limited to the following information:

(a) Title, number, and date of this specification.

(b) Type and size of container desired.

(c) Place of inspection

(d) Minimum lot size, if applicable.

(e) Place of delivery

(f) Request for copies of inspection data, if source inspected.

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(g) Selection of applicable levels of packing and packaging (see 5.1 and 5.2)

(h) Marking requirements (see 5.3)

(i) Data required (see 3.1 and 4.5.2.5)

6.3 Conflicting Requirements. -Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred to in writing to the procuring activity or appointed agent for interpretation and clarification.

6.4 Request for Deviation. -Request for deviation from this or referenced specifications, materials, applicable drawings, or processes shall be forwarded to the procuring activity prior to manufacture of the affected item or items. All deviations shall be limited to the contract under which they were granted.

6.5 Request for Waiver. -Requests for waiver from this specification, applicable drawings, specifications, materials, or processes shall be forwarded to the procuring activity. Waivers shall be limited to the specific units for which they are granted.

(a) Exact nature of the defect

(b) CD number and defect classification, if any

(c) Quantity of items involved

6.6 Value Engineering. -Manufacturers of material covered by this specification are encouraged to submit value engineering suggestions to the procuring activity in an effort to reduce cost.

6.7 Safety Precautions. -This is a potentially dangerous material. All detailed safety precautions shall be strictly observed as set forth by the manufacturer and user.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the spec-
on in procurement of products for ultimate use by the Department of Defense. This sheet is provided for ob-
; information on the use of this specification which will insure that suitable products can be procured with a
) amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on
)n reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

ICATION

,-G-23949(Wep) GRAINS, TUBULAR, PYROTECHNIC

ZATION *(of submitter)* | CITY AND STATE

CT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

AL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

MENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

MARKS *(Attach any pertinent data which may be of use in improving this specification. If there are addi-
onal papers, attach to form and place both in an envelope addressed to preparing activity)*

TTED BY *(Printed or typed name and activity)* | DATE

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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